

Roll No: _____

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION
Question Paper

Month and year: August 2026

Program: **B.Tech**

Subject: Artificial Intelligence and Soft Computing

Maximum Marks: 50

Number of Questions to be attempted: 5

Total no. of pages used: 1

Semester: 4th

Course code: CSPC 204/ITPC 204

(CS,IT, M&C)

Time allowed: 03 Hours

Total No of Questions: 5

Note: Unless stated otherwise, the symbols have their usual meanings in context with subject. Assume suitably and state, additional data required, if any. The Candidates, before starting to write the solutions, should please check the question Paper for any discrepancy, and also ensure that you have been delivered the question paper of right **course no.** and right **subject title.**

Q1.		
a)	Define soft computing. Differentiate between soft computing and hard computing techniques.	5
b)	Does any heuristic satisfying the admissibility constraint, automatically ensure consistency constraints of A star. Justify your answer with an example.	5
Q-2.		
a)	Formulate a solution for water jug problem using the simulated annealing algorithm. Explain the cooling and iteration procedure.	6
b)	What are uninformed search techniques? Explain the difference between breadth first search and uniform cost search techniques. Under what circumstances do they behave similarly.	4
Q-3.		
a)	Explain the minimum conflict algorithmic approach for solving constraint satisfaction problems. How is it different from backtracking approach?	5
b)	Explain the AND OR search algorithm with an example.	5
Q-4.		
a)	Demonstrate the working of resolution algorithm for propositional logic with an example. Does it offer completeness.	5
b)	Explain the following terms: i) ISA relation, ii) Facets, iii) Reification, iv) Abduction, v) Frames	5
Q-5.		
a)	Formulate ant colony optimization-based solution for water jug problem. Also elaborate the trail update, pheromone vaporization and decay procedures for the solution.	5
b)	Demonstrate the functioning of artificial bee colony algorithm with an example.	5